significance—reduces the uncertainty of predicting an occurrence in the sense world to the uncertainty in translating that occurrence from the sense world into the world picture and in retranslating it from the world picture into the sense world.

Classical physics was but little concerned with this uncertainty; its main concern was to follow through the causal point of view in the consideration of the occurrences in the world picture, and this was where it achieved its great results. Specifically, it succeeded also in finding a satisfactory interpretation, on the basis of a strict causality, for the above mentioned irregular vibratory phenomena corresponding to the Brownian molecular movement. The indeterminists see no real problem here. For since they look for irregularity behind every rule, statistical regularity is that which gives them direct satisfaction. Therefore, they are satisfied also with the assumption that the collision of two individual molecules, as well as the impact of the molecules against the wall of the vessel, occurs solely according to statistical laws. Nevertheless, this assumption is as little justified as would be the conclusion that the charge of an individual electron is located on its surface just because in a charged conductor the electrons are all located on its surface. On the other hand, the determinists who, on the contrary, look for a rule behind every irregularity, were led to the problem of basing a theory of the gas laws on the premise that the collision of two individual molecules is determined in a strict causal manner. The solution of this problem was the life work of Ludwig Boltzmann, and it represents one of the most beautiful triumphs of theoretic research. For it not only yields the principle, confirmed by actual measurements, that the mean energy of the oscillations about the state of equilibrium is proportional to the absolute temperature, but it permits also a remarkably accurate computation of the absolute number and mass of the molecules, based on a measurement of these oscillations, as in the case of a highly sensitive torsion balance.

Such outstanding achievements seemed to justify the hope that the world picture of classical physics would in principle accomplish its task, and that a steady improvement and refinement of the technique and methods of measurement would progressively reduce the significance of the uncertainties accompanying the translation into and from the sense world. But the introduction of the elementary quantum of action destroyed this hope at one blow and for good.

The so-called Uncertainty Principle, discovered and formulated by Heisenberg, constitutes a characteristic feature of quantum mechanics; it asserts that for any two canonically conjugate magnitudes, such as position and momentum or time and energy, only one can be measured to any desired degree of accuracy, so that an increase in the precision of the measurement of one magnitude is accompanied by a proportional decrease in the precision of